Number System

09 June 2026 05:08

How are numeric data items actually stored in the computer memory?

How much space(memory locations) is allocated for each type of data?

- byte, short, int, long, float, double, char, boolean.

How are characters and strings are stored in memory?

Number System:: The basics

We are accustomed to using the so called **decimal number system.**

- Ten digits:: 0,1,2,3,4,5,6,7,8,9
- Every digit position has a weight which is a power of 10.
- Base or radix is 10

Example:

$234 = 2 * 10^2 + 3 * 10^1 + 4 * 10^0$

$250.67 = 2 * 10^2 + 5 * 10^1 + 0 * 10^0 + 6 * 10^{-1} + 7 * 10^{-2}$

Binary Number System

Two digits:

- **0 and 1**
- **Every digit position has a weight which is a power of 2.**
- **Base or radix is 2**

Example:

$110 = 1 * 2^2 + 1 * 2^1 + 0 * 2^0$

$110.01 = 1 * 2^2 + 1 * 2^1 + 0 * 2^0 + 0 * 2^{-1} + 1 * 2^{-2}$

Binary to decimal conversion

Each digit position of a binary number has a weight.

- **Some power of two**

A binary number:

$B = b_{n-1}b_{n-2}\ldots b_1 b_0 b_{-1} b_{-2}\ldots b_{-m}$

Corresponding value in decimal:

$$D = \sum_{i=-m}^{n-1} b_i \cdot 2^i$$

Examples:

$10101_2 = 1\times2^5 + 0\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0$

$= 32 + 0 + 8 + 0 + 2 + 1$

$= 43_{10}$

$= 43_{10}$

$0.0101_2 = 0\times2^{-1} + 1\times2^{-2} + 0\times2^{-3} + 1\times2^{-4}$

$= 0.2500 + 0.0625$

$= 0.3125_{10}$

$101.11 = 5 + 1\times2^{-1} + 1\times2^{-2}$

$5 + 0.50 + 0.25 = 5.75_{10}$

Decimal-to-Binary Conversion

Consider the integer and fractional parts separately

For integer part,

- **Repeatedly divide the given number by two and go on accumulating the remainder until the number becomes zero.**
- **Arrange the reminders in reverse order.**

For the fractional part:

- **Repeatedly multiply the given fraction by two.**
  - **Accumulate the integer part (0, 1).**
  - **If the integer part is one, chop it off.**
- **Arrange the integer part in order they are obtained.**

Examples:

$(239.25)_{10} = (\ ?\ )_2$

2 | 239		
2 | 119	1	↑
2 | 59	1	
2 | 29	1	
2 | 14	1	
2 | 7	0	
2 | 3	1	
2 | 1	1	
0	1	

365

$11101101_2 = 949_{10}$

$(239)_{10} = (11101111)_2$

2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1024	512	256	128	64	32	16	8	4	2	1
			1	1	1	0	1	1	1	1
		1	0	1	1	0	1	1	0	1
	1	1	1	0	1	1	0	1	0	1
			1	1	1	0	1	1	1	0

1023
0 to 511

1088

0.25×2

$0 \mid 50 \times 2$

$1 \mid 00$

$239 + 0.25 = 11101111 + .01$

$(239.25)_{10} = (11101111.01)_2$

$1101110_2 = 110$

$\rightarrow 1101_{10} = (44d)_{12}$

Python code for conversion of numbers

```
bs = '1101'
print(int(bs,2)) ← binary to int
13
print(oct(13)) ← int to octal
0o15
print(hex(13)) ← int to hex
0xd
print(bin(13)) ← int to binary
0b1101
```

Java

```
public class Coverersion
{
    public static void convertIntegerToBinary(int v){
        System.out.print("\nBinary equivalent of " + v +" is: " +
        Integer.toBinaryString(v));
        System.out.print("\n?: "+Integer.toUnsignedString(110, 2));

        System.out.print("\n ??: "+Integer.toString(1101, 16));
    }
}
```

int is converted to binary $(1101110)_2$

Methods available in Java for conversion:

```
String   toBinaryString(int)
String   toHexString(int)
String   toOctalString(int)
String   toString(int)
String   toString(int, int)
long     toUnsignedLong(int)
String   toUnsignedString(int)
String   toUnsignedString(int, int)
```

```
Integer.to
```

Hexadecimal Number System
A compact way of representing binary numbers
16 different symbols(radix = 16)

$16 = 2^4$ ← bits

0 0 0 0	0
0 0 0 1	1
0 0 1 0	2
0 0 1 1	3
0 1 0 0	4
0 1 0 1	5
0 1 1 0	6
0 1 1 1	7
1 0 0 0	8
1 0 0 1	9

0	1	1	1	7	
1	0	0	0	8	
1	0	0	1	9	
1	0	1	0	A	(10)
1	0	1	1	B	(11)
1	1	0	0	C	(12)
1	1	0	1	D	(13)
1	1	1	0	E	(14)
1	1	1	1	F	(15)

Binary to Hexadecimal Conversion

For the integer part:

- **Scan the binary number from right to left.**
- **Translate each group of four bits into the corresponding hexadecimal digit.**
  - **Add leading zeros if necessary.**

For the fractional part:

- **Scan the binary number from left to right.**
- **Translate each group of four bits into the corresponding hexadecimal digit.**
  - **Add trailing zeros if necessary.**

For example:-

$(0\,1011101.01101110)_2 = (2ed.67)_{16}$

2 e d . 6 7

$(0.10000100)_2 = (0.84)_{16}$

8	4	2	1
1	1	0	1
1	1	1	0
1	0	0	0
0	1	0	1
1	0	1	0
		1	1
1	1	0	1
0	0	1	1
0	0	1	0

Hexadecimal to binary Conversion

Translate every hexadecimal digit into its 4 bit binary equivalent.

$(3A5)_{16} = (1110100101)_2$

$(12.3D)_{16} = (10010.00111101)_2$